Data Science and Visualization

Unit I

FOUNDATIONS OF DATA SCIENCE

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Unit I – Foundations of Data Science

Defining Data Science - The Data Science Process - Data Types and Attributes - Python Essentials: NumPy & Pandas - Data Loading and Inspection - Data Visualization Goals - Introduction to Matplotlib & Seaborn - Basic Plotting and Customization.

Python Essentials

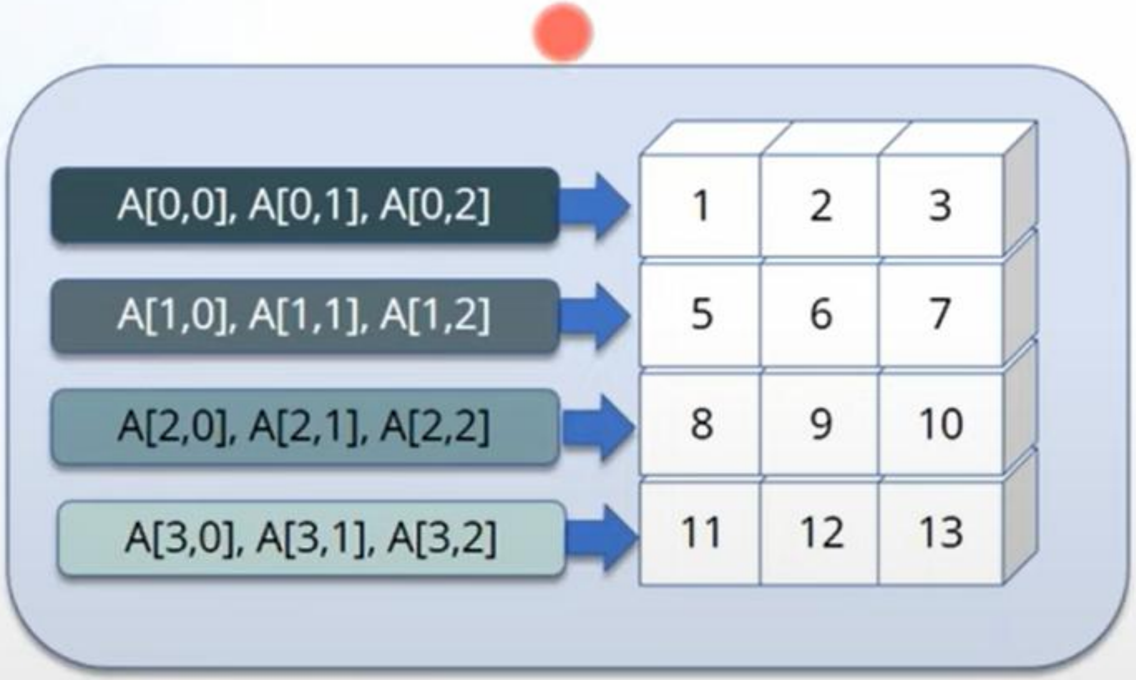
NumPy

Pandas

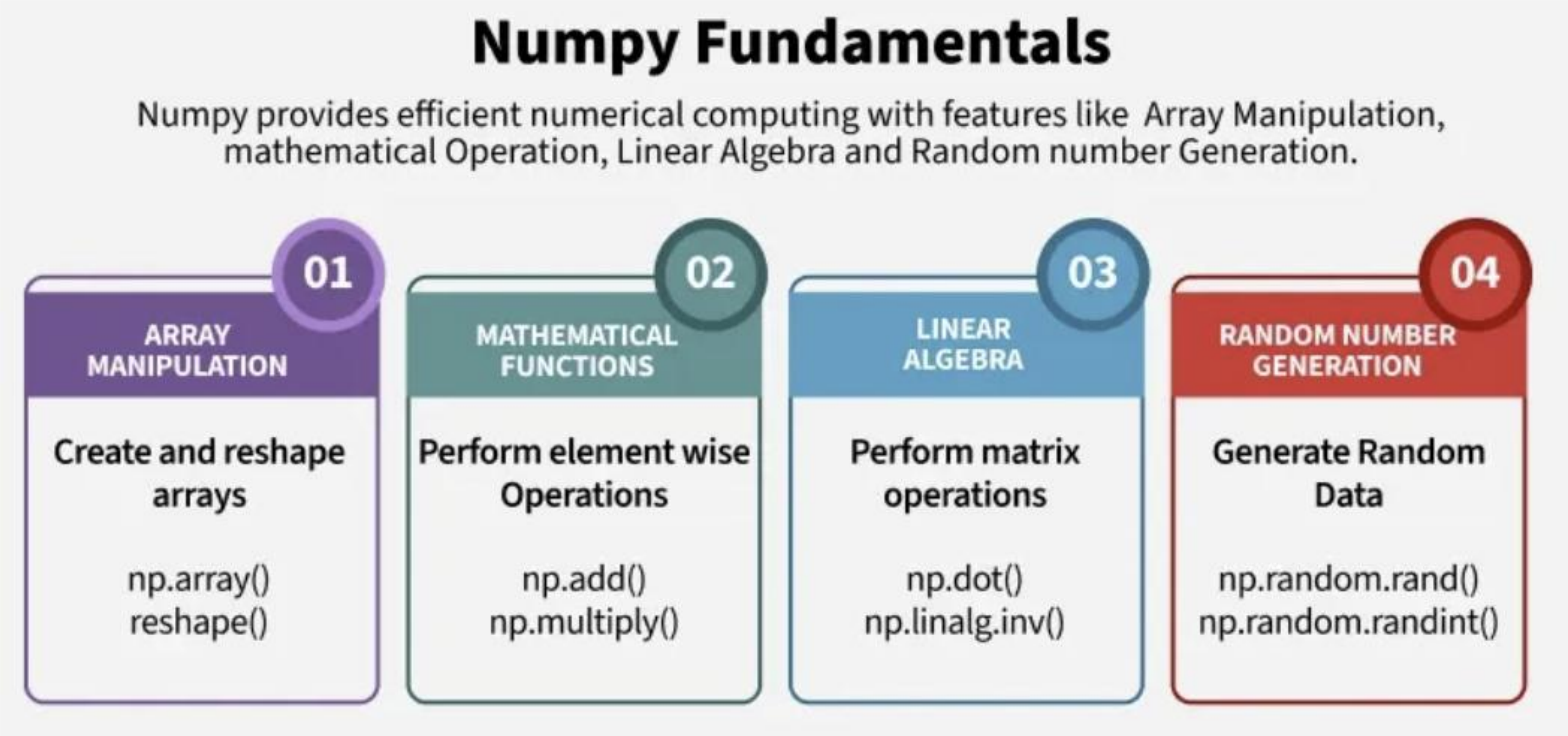
	NumPy	Pandas
Basics	For numerical calculations, support for arrays	For data manipulation and analysis
Good to know	np.array([]), np.info	pd.read_csv(), df.drop(), df.plot(kind='line')
Key Functions	Array creation, indexing, slicing; Mathematical operations on arrays	Data import & export to/from var. formats, data filtering & aggregation, Data Exploration

NumPy

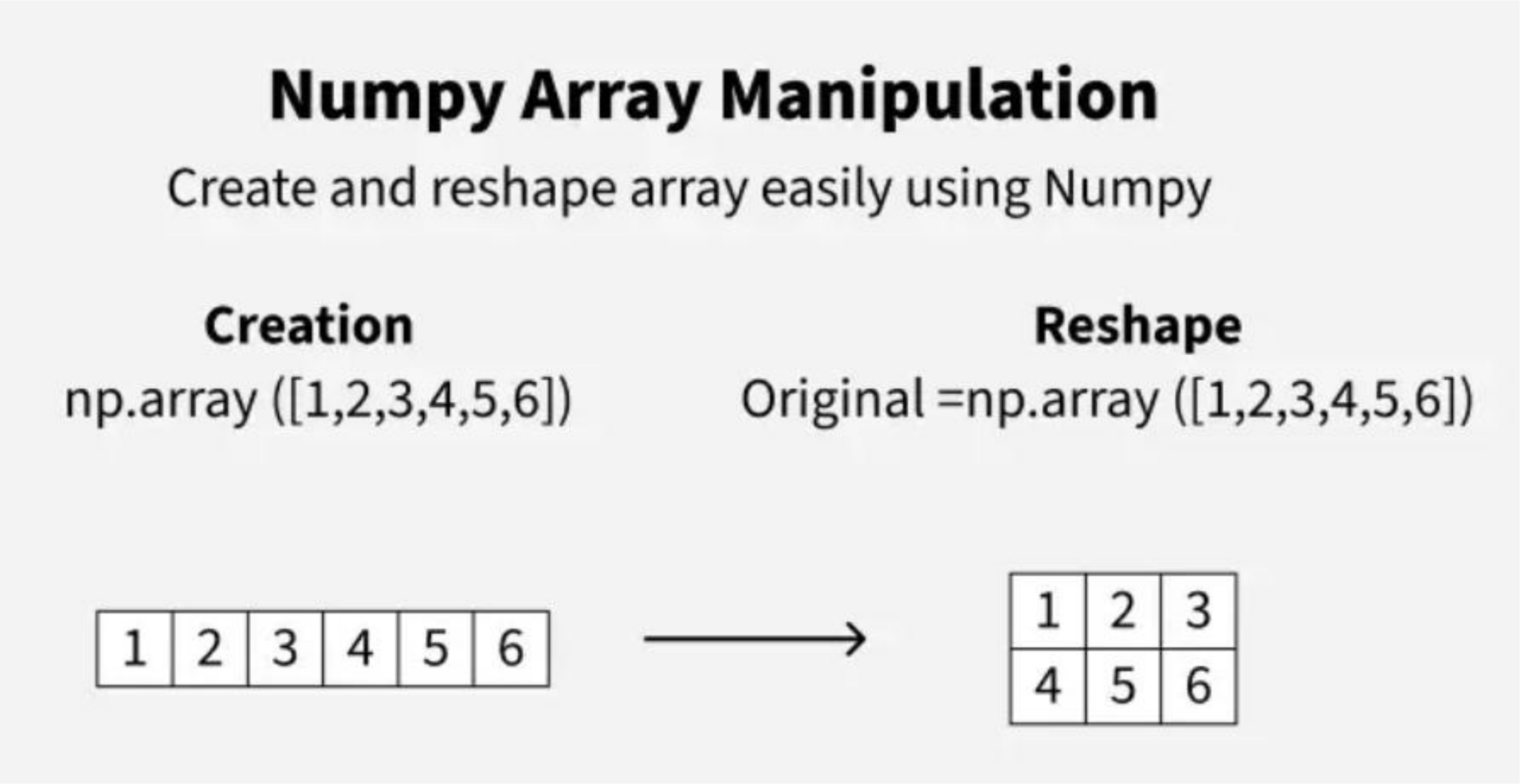
- Numpy is the core library for scientific computing in Python.
- It provides a high-performance multidimensional array object, and tools for working with these arrays.



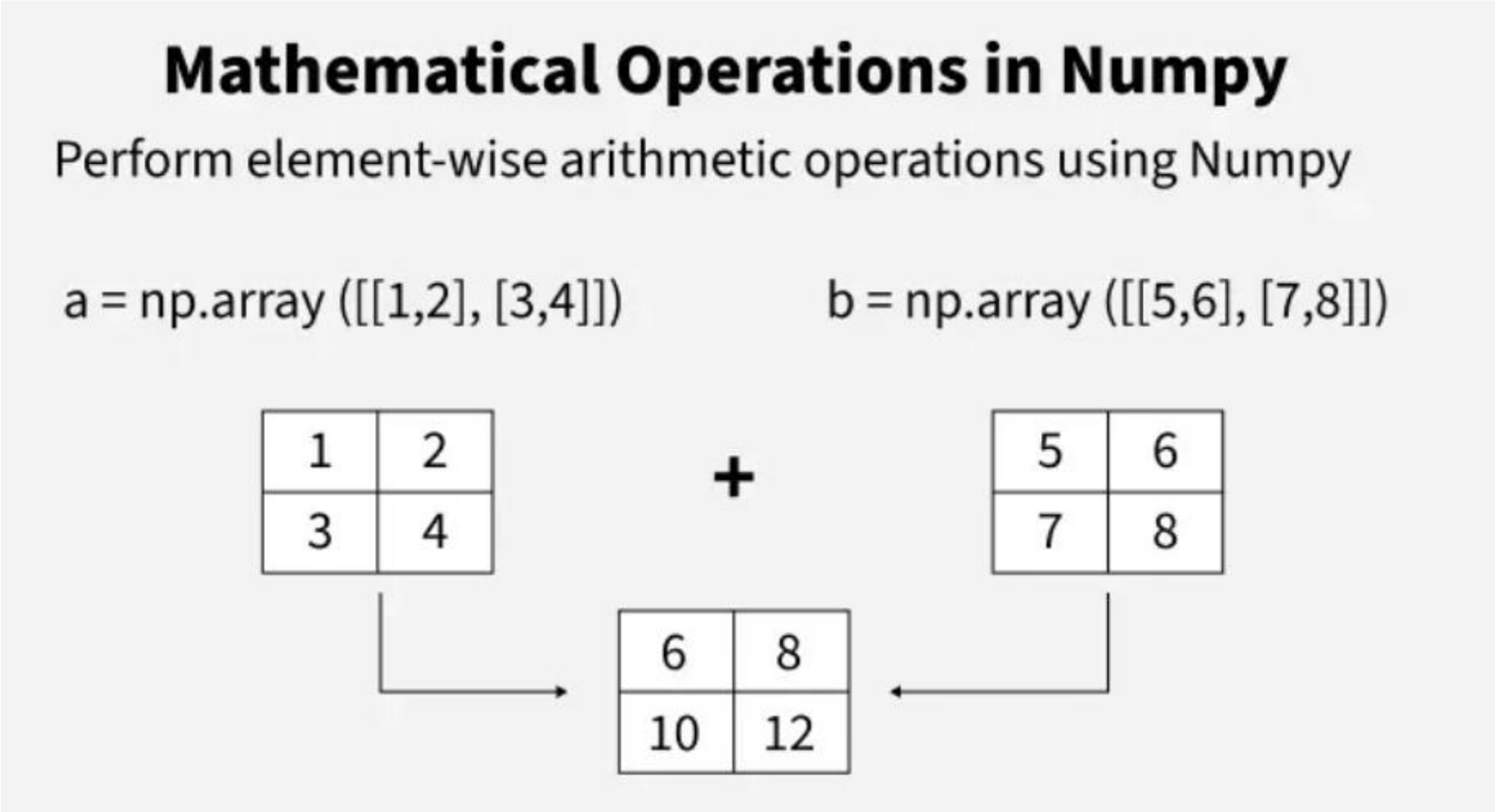
NumPy


Numpy Fundamentals
Numpy provides efficient numerical computing with features like Array Manipulation, mathematical Operation, Linear Algebra and Random number Generation.
01
ARRAY MANIPULATION
Create and reshape arrays
np.array()
reshape()
02
MATHEMATICAL FUNCTIONS
Perform element wise Operations
np.add()
np.multiply()
03
LINEAR ALGEBRA
Perform matrix operations
np.dot()
np.linalg.inv()
04
RANDOM NUMBER GENERATION
Generate Random Data
np.random.rand()
np.random.randint()

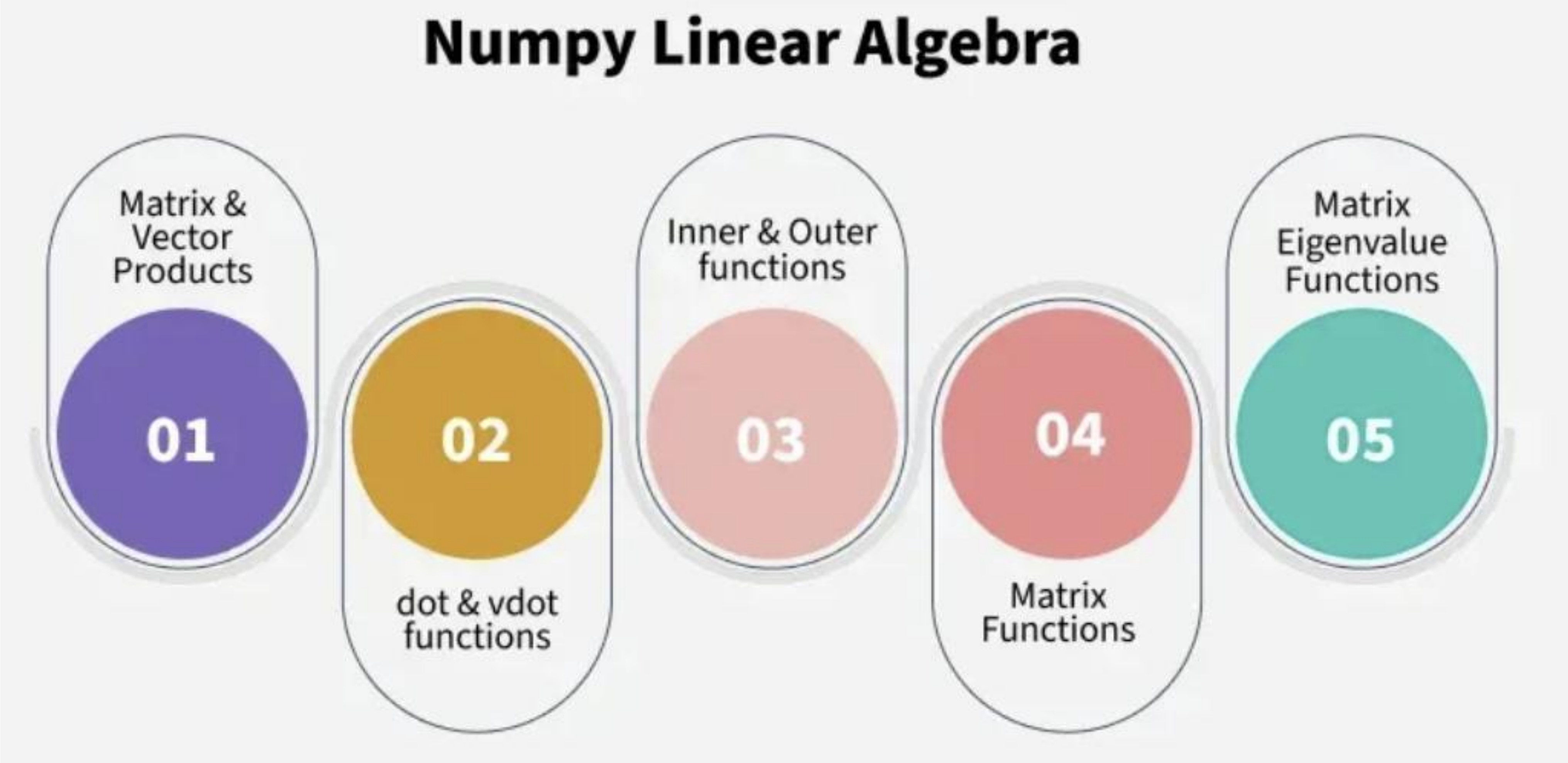
NumPy


Numpy Array Manipulation
Create and reshape array easily using Numpy
Creation
np.array ([1,2,3,4,5,6])
Reshape
Original =np.array ([1,2,3,4,5,6])
1 2 3 4 5 6
1 2 3
4 5 6

NumPy


Mathematical Operations in Numpy
Perform element-wise arithmetic operations using Numpy
a = np.array ([[1,2], [3,4]])
b = np.array ([[5,6], [7,8]])
1 2
3 4
+
5 6
7 8
6 8
10 12

NumPy



Numpy Linear Algebra
Matrix & Vector Products
01
02
dot & vdot functions
Inner & Outer functions
03
04
Matrix Functions
Matrix Eigenvalue Functions
05

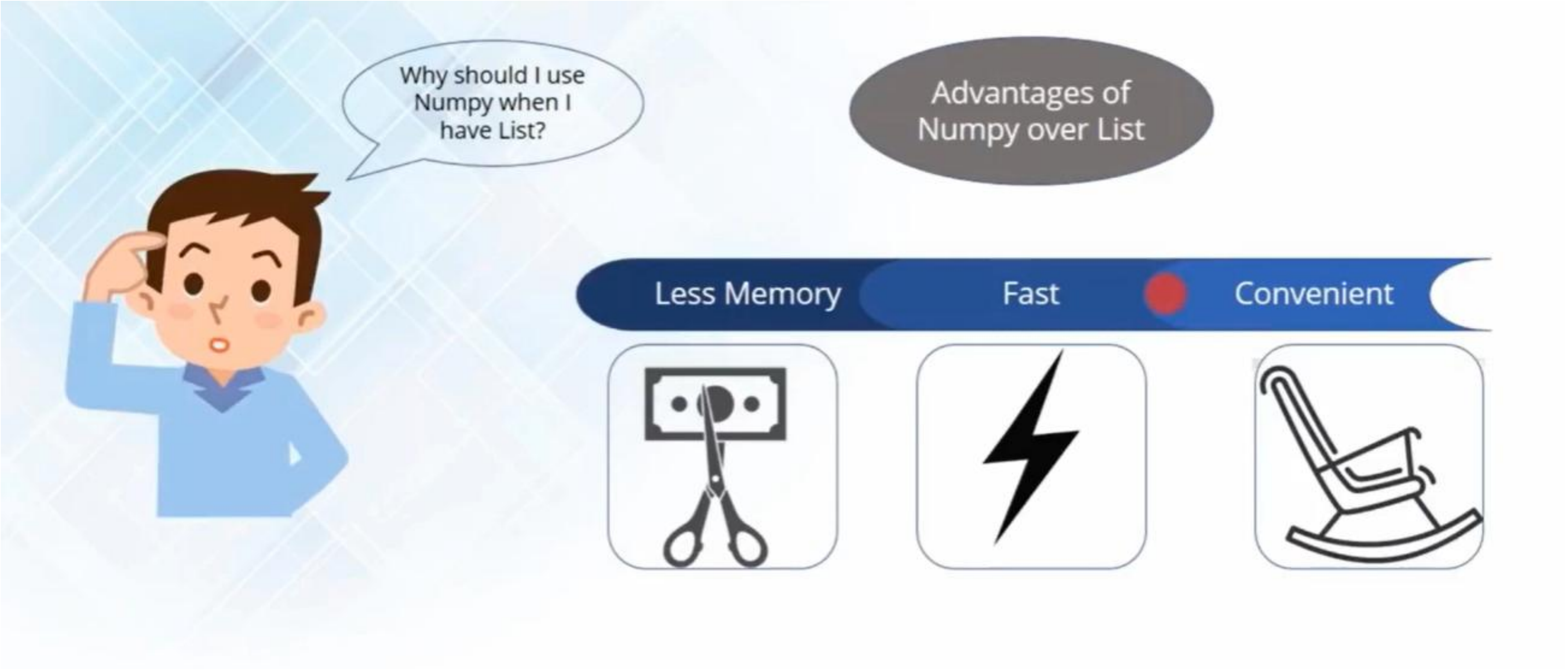
NumPy Array

```
import numpy as np
a=np.array([(1,2,3),(4,5,6)])
print(a)
```

[1] ✓ 0.4s

```
[[1 2 3]
 [4 5 6]]
```

NumPy Array


Why should I use Numpy when I have List?
Advantages of Numpy over List
Less Memory
Fast
Convenient

NumPy - Occupies less memory

```
import time
import sys
s=range(1000)
print(sys.getsizeof(5)*len(s))
D=np.array(10000)
print(D.size*D.itemsize)
```

[7] ✓ 0.0s

```
28000
4
```

NumPy - Fast

```
SIZE=100000

L1=range(SIZE)
L2=range(SIZE)

A1=np.array(SIZE)
A2=np.array(SIZE)

start= time.time()

result=[(x,y) for x,y in zip(L1,L2)]
print((time.time()-start)*1000)#list compute time

start= time.time()
result=A1+A2
print((time.time()-start)*1000)#Numpy Compute list
```

[14] ✓ 0.5s

```
515.4762268066406
2.992391586303711
```

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NumPy Operations

```python
# Numpy operations
a=np.array([(1,2,3),(4,5,6)])
print(a.itemsize)# Finding byte size of each element

b=np.array([(1,2,3,4,6,7,8),(4,5,6,9,11,3,8)])
print(b.shape)#Finding the size of an array

c=np.array([(1,2,3,9),(4,5,6,10)])
print(c)
d=c.reshape(4,2)#transpose matrix/Reshape
print(d)

e=np.array([(1,2,3,9),(4,5,6,10),(7,8,9,10)])
print(e[0:,2])#Slice


f=np.array([(1,2,3,9),(4,5,6,10),(7,8,9,10)])
print(f[0:2,3])#Slice
```

[39] ✓ 0.0s

```
4
(2, 7)
[[ 1  2  3  9]
 [ 4  5  6 10]]
[[ 1  2]
 [ 3  9]
 [ 4  5]
 [ 6 10]]
[3 6 9]
[ 9 10]
```

https://scipy-lectures.org/intro/numpy/operations.html

```python
a=np.linspace(1,3,9)
print(a)
```

[43] ✓ 0.0s

```
[1.   1.25 1.5  1.75 2.   2.25 2.5  2.75 3.  ]
```

NumPy Operations

01 ndim

02 itemsize()

03 dtype()

04 reshape()

05 slicing()

06 linspace()

07 min/max()

08 sum()

09 sqrt()/std()

10 hstack/vstack()

pandas

What is Pandas in Python?

Pandas is used for data manipulation, analysis and cleaning.

Pandas

	Bounce rate	Day	Visitors
0	20	1	1000
1	20	2	700
2	23	3	6000
3	15	4	1000
4	10	5	400
5	34	6	350

Slicing the starting 2 rows

	Bounce rate	Day	Visitors
0	20	1	1000
1	20	2	700

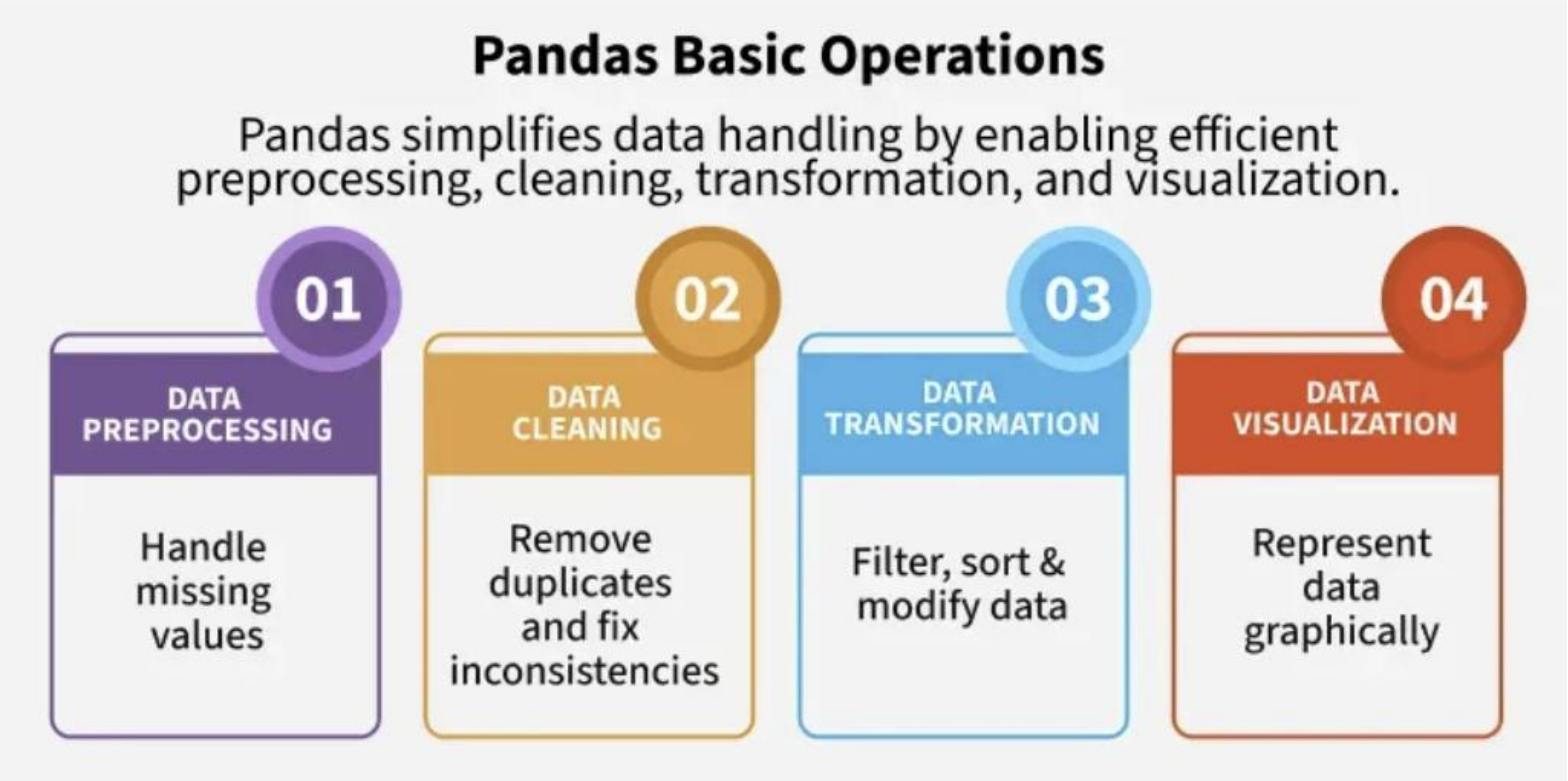
Slicing the last 2 rows

	Bounce rate	Day	Visitors
4	10	5	400
5	34	6	350

What are DataFrames and Series?

A DataFrame is A Two-dimensional, size-mutable, potentially heterogeneous tabular data.

Pandas


Pandas Basic Operations
Pandas simplifies data handling by enabling efficient preprocessing, cleaning, transformation, and visualization.
01
DATA PREPROCESSING
Handle missing values
02
DATA CLEANING
Remove duplicates and fix inconsistencies
03
DATA TRANSFORMATION
Filter, sort & modify data
04
DATA VISUALIZATION
Represent data graphically

Pandas

Data Preprocessing

Handle missing values and clean raw data for analysis.
Before (Table with Missing Values

Before (Table with Missing Values)

Name	Age	Salary_Amount
Aryan	25	NaN
Harsh	NaN	5000
Kunal	30	7000

After (Missing Values Filled)

Name	Age	Salary
Aryan	25	0
Harsh	28	5000
Kunal	30	7000

Operation Applied: Filled missing values with 0 / average using fillna()

Pandas

Data Cleaning

Remove duplicates and fix column inconsistencies.
Before (Duplicate Rows & Bad Column Names):

Before

Name	Age	Salary_Amount
Aryan	25	5000
Harsh	30	7000
Aryan	25	5000

After

Name	Age	Salary
Aryan	25	5000
Harsh	30	7000

Operation Applied: Removed duplicates using drop_duplicates(), renamed columns using rename()

Pandas

Data Transformation

Filter and sort data for better insights.

Before (unfiltered Data)

Product	Category	Price
Laptop	Electronics	800
Hoodie	Clothing	20
Smartphone	Electronics	500
Jeans	Clothing	40

After (Filtered & Sorted Data)

Product	Category	Price
Smartphone	Electronics	500
Laptop	Electronics	800

Operation Applied: Filtered Electronics using query(), sorted by price using sort_values()

Pandas

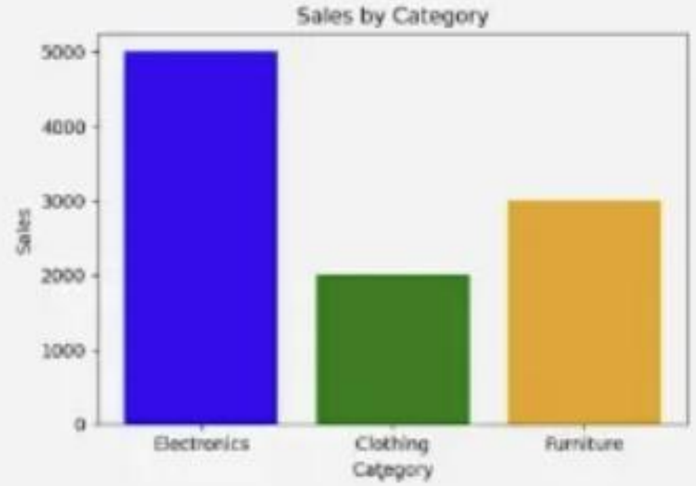
Data Visualization

Represent data graphically for insights

Before (Raw Data Table)

Category	Price
Electronics	5000
Clothing	2000
Furniture	3000

After (Bar Chart Visual)



Operation Applied: Plotted sales data using matplotlib

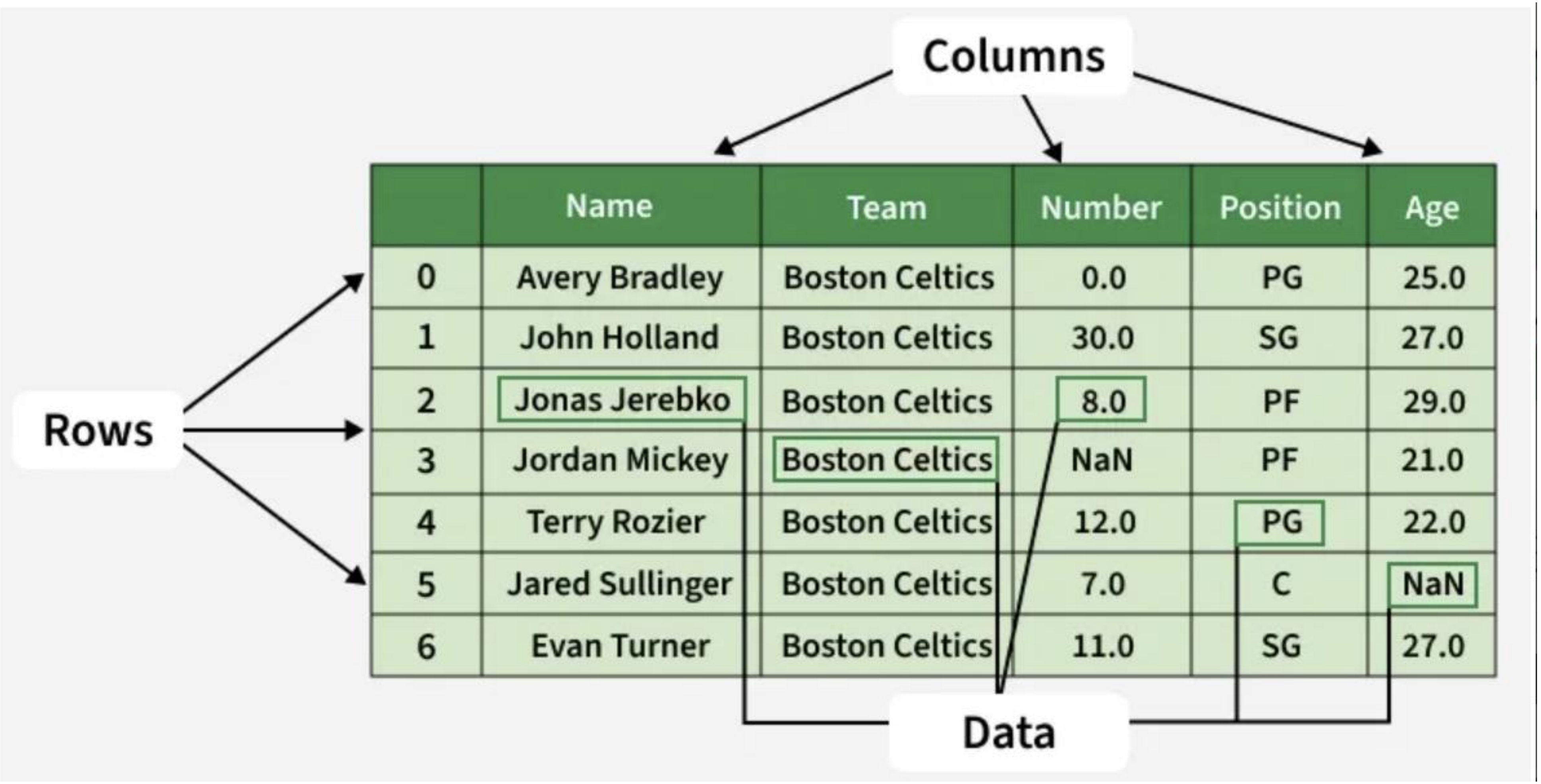
Pandas Data Frames

- A Pandas DataFrame is a **two-dimensional table-like structure** in Python where data is arranged in rows and columns.
- It's one of the most commonly used tools for handling data and makes it easy to organize, analyze and manipulate data.
- It can store different types of data such as numbers, text and dates across its columns.

The main parts of a DataFrame are:

- ✓ **Data:** Actual values in the table.
- ✓ **Rows:** Labels that identify each row.
- ✓ **Columns:** Labels that define each data category.

Pandas Data Frames



	Name	Team	Number	Position	Age
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

https://www.geeksforgeeks.org/pandas/python-pandas-dataframe/

Pandas Data Frames

Example: Create DataFrame using dictionary:

```
import pandas as pd

data = {'Name': ['Alice', 'Bob', 'Charlie'],'Age': [25, 30, 35]}
df = pd.DataFrame(data)
print(df)
```

Output

	Name	Age
0	Alice	25
1	Bob	30
2	Charlie	35

Pandas Data Frames

```python
import pandas as pd
df = pd.read_csv("people.csv")
print(df)
```

	First Name	Last Name	Sex	Email	Date of birth	Job Title
0	Shelby	Terrell	Male	elijah57@example.net	1945-10-26	Games developer
1	Phillip	Summers	Female	bethany14@example.com	1910-03-24	Phytotherapist
2	Kristine	Travis	Male	bthompson@example.com	1992-07-02	Homeopath
3	Yesenia	Martinez	Male	kaitlinkaiser@example.com	2017-08-03	Market researcher
4	Lori	Todd	Male	buchananmanuel@example.net	1938-12-01	Veterinary surgeon

Pandas Data Frames

```python
import pandas as pd

data = {'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva'],
        'Age': [24, 27, 22, 32, 29],
        'City': ['New York', 'Los Angeles', 'Chicago', 'Houston', 'Phoenix']}

df = pd.DataFrame(data)

# View the first 3 rows of the DataFrame
print("First 3 rows using head():")
print(df.head(3))

# View the last 2 rows of the DataFrame
print("\nLast 2 rows using tail():")
print(df.tail(2))

# Get a concise summary of the DataFrame
print("\nDataFrame summary using info():")
df.info()
```

First 3 rows using head():

	Name	Age	City
0	Alice	24	New York
1	Bob	27	Los Angeles
2	Charlie	22	Chicago

Last 2 rows using tail():

	Name	Age	City
3	David	32	Houston
4	Eva	29	Phoenix

```
DataFrame summary using info():
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   Name    5 non-null      object
 1   Age     5 non-null      int64 
 2   City    5 non-null      object
dtypes: int64(1), object(2)
memory usage: 248.0+ bytes
```

Pandas Data Frames

Example : Basic Indexing (Selecting a single column) with us

```
import pandas as pd

data = {'Name': ['Alice', 'Bob', 'Charlie'],'Age': [25, 30, 35]}
df = pd.DataFrame(data)

# Select a single column
age_column = df['Age']
print(age_column)
```

```
0     25
1     30
2     35
Name: Age, dtype: int64
```

Pandas Data Frames

```
import pandas as pd

data = {'Name': ['Alice', 'Bob', 'Charlie'], 'Age': [25, 30, 35]}
df = pd.DataFrame(data)

# Using .loc[] to select rows by label
row_by_label = df.loc[1]  # Selects the row with index label 1 (Bob's data)

# Using .iloc[] to select rows by position
row_by_position = df.iloc[1]  # Selects the second row (Bob's data)

print("Row by label:\n", row_by_label)
print("Row by position:\n", row_by_position)
```

✓ 0.1s

```
Row by label:
 Name    Bob
Age      30
Name: 1, dtype: object
Row by position:
 Name    Bob
Age      30
Name: 1, dtype: object
```

NumPy Vs Pandas

	NumPy	Pandas
Focus	Numerical computations	Tabular data analysis
Goal	Fast math on arrays	Easy data manipulation
Tools	ndarray (multi-dimensional array)	DataFrame (like Excel)
Skills	Math ops, vectorization, reshaping	Cleaning, filtering, grouping
Beginner Tip	Learn first to understand data structures	Learn next for real-world projects

References

NumPy

https://www.w3schools.com/python/numpy/numpy_intro.asp

https://www.geeksforgeeks.org/python/numpy-tutorial/

https://numpy.org/

Pandas

https://www.w3schools.com/python/pandas/default.asp

https://www.geeksforgeeks.org/pandas/introduction-to-pandas-in-python/

https://pandas.pydata.org/

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